

In [1]:

```
import numpy as np
import pandas as pd
```

Creating Series from list

In [2]:

```
lst = [12,13,14,15,16]
arr = np.array(lst)
s = pd.Series(lst)
```

In [3]:

```
print(arr)
```

```
[12 13 14 15 16]
```

In [4]:

```
print(s)
```

```
0    12
1    13
2    14
3    15
4    16
dtype: int64
```

In [7]:

```
s1 = pd.Series(['hello',1,2,3,4,5])
s1
```

Out[7]:

```
0    hello
1         1
2         2
3         3
4         4
5         5
dtype: object
```

In [8]:

```
s2 = pd.Series([1.2,1,2,3,4,5,6])  
s2
```

Out[8]:

```
0    1.2  
1    1.0  
2    2.0  
3    3.0  
4    4.0  
5    5.0  
6    6.0  
dtype: float64
```

Creating Series from Numpy Array

In [9]:

```
lst1 = [1,2,3,2,4,2,4,6,3,76,5,54,34]  
arr1 = np.array(lst1)
```

In [10]:

```
arr1
```

Out[10]:

```
array([ 1,  2,  3,  2,  4,  2,  4,  6,  3, 76,  5, 54, 34])
```

In [11]:

```
s3 = pd.Series(arr1)  
s3
```

Out[11]:

```
0     1  
1     2  
2     3  
3     2  
4     4  
5     2  
6     4  
7     6  
8     3  
9    76  
10    5  
11   54  
12   34  
dtype: int32
```

Changing Series Index

In [13]:

```
s4 = pd.Series(['hello', 'world', 'python'], index=[3,2,1])
s4
```

Out[13]:

```
3    hello
2    world
1    python
dtype: object
```

In [15]:

```
s5 = pd.Series([1, 'Rahul', 'BCA', '78', 'B+'], index=['Rollno', 'Name', 'Class', 'Marks', 'Grade'])
s5
```

Out[15]:

```
Rollno    1
Name      Rahul
Class     BCA
Marks     78
Grade     B+
dtype: object
```

In [17]:

```
# Length of index list must be same as length of the Series.
```

Creating Series from Dictionary

In [18]:

```
d = {'Name': 'Manav', 'Class': 'Btech', 'Marks': 50, 'Grade': 'C'}

s7 = pd.Series(d)
s7
```

Out[18]:

```
Name      Manav
Class     Btech
Marks      50
Grade      C
dtype: object
```

In [19]:

```
d = [{ 'Name': 'Manav', 'Class': 'Btech', 'Marks': 50, 'Grade': 'C' },
      { 'Name': 'Manav', 'Class': 'Btech', 'Marks': 50, 'Grade': 'C' }
    ]

s8 = pd.Series(d)
s8
```

Out[19]:

```
0    { 'Name': 'Manav', 'Class': 'Btech', 'Marks': 5...
1    { 'Name': 'Manav', 'Class': 'Btech', 'Marks': 5...
dtype: object
```

Iterating over Series

In [20]:

```
s9 = pd.Series([1,2,3,2,4,2,4,6,3,76,5,54,34])
s9
```

Out[20]:

```
0      1
1      2
2      3
3      2
4      4
5      2
6      4
7      6
8      3
9     76
10     5
11    54
12    34
dtype: int64
```

In [21]:

```
for i in range(0, len(s9)):
    print(s9[i])
```

```
1
2
3
2
4
2
4
6
3
76
5
54
34
```

In [22]:

```
for i in s9:  
    print(i)
```

```
1  
2  
3  
2  
4  
2  
4  
6  
3  
76  
5  
54  
34
```

Accessing the Series Elements

In [23]:

```
d1 = {'Name': 'Manav', 'Class': 'Btech', 'Marks': 50, 'Grade': 'C'}  
  
s11 = pd.Series(d1)  
s11
```

Out[23]:

```
Name      Manav  
Class      Btech  
Marks       50  
Grade       C  
dtype: object
```

In [24]:

```
s11['Name']
```

Out[24]:

```
'Manav'
```

In [25]:

```
s11[0]
```

Out[25]:

```
'Manav'
```

In [26]:

```
s11[0] = 'Vikas'
```

In [27]:

```
s11
```

Out[27]:

```
Name      Vikas
Class     Btech
Marks      50
Grade      C
dtype: object
```

In []: